Stakeholder Question Catalogue 05

Lecturers and Tutors

80 teaching-decision questions mapped to proportionate sensing, authoritative institutional sources, explicit operation types, evidence quality, uncertainty, permissions and safe answer boundaries.

STAKEHOLDER	Lecturers, tutors, seminar leaders, demonstrators and other teaching staff using Abacws for lectures, seminars, tutorials, practicals, office hours, hybrid teaching, visiting speakers and irregular teaching schedules.
QUESTION RANGE	Session preparation and delivery; room suitability, accessibility and comfort; AV, hybrid and recorded teaching; transitions and office hours; visitors and events; inclusive teaching, safety, sustainability, service reliability and evidence transparency.
EVIDENCE MODEL	Proposed, correctly located sensing is combined with authorised timetable, room, booking, access, accessibility, AV/IT, BMS, energy, maintenance, visitor, policy and safety sources. Every record states minimum evidence, operation type, source precedence, quality threshold, uncertainty and safe failure.

Core principle: Support lecturer-owned teaching decisions with verified evidence. Never monitor teaching content, identify or track people, infer attendance or performance, bypass responsible authorities, or express more certainty than the evidence permits.

1. Catalogue scope and design principles

This catalogue focuses on decisions that lecturers, tutors, seminar leaders and demonstrators may reasonably want a conversational university building to support before, during and after teaching. It does not transfer responsibilities from timetabling, room booking, AV/IT, facilities, accessibility, security or safety authorities.

Proposed capability, not current fact

Every sensor, integration and service is a design requirement unless deployment, room coverage, permissions and operational status are independently verified. The catalogue does not claim that all capabilities currently exist.

Current building information

The May 2021 floor plan is a historical spatial reference. Current room use, ownership, capacity, staff access, bookability, specialist facilities, equipment, routes, lifts and services require authoritative verification.

Lecturer-owned decisions

Questions support preparation, delivery, pacing, route choice, accessibility communication, office-hour planning, contingency and evidence-based requests. The service does not move classes, book rooms, change controls or operate building systems.

No teaching surveillance

Do not inspect teaching materials, recordings, screens, speech, conversations, network traffic or device content. Do not identify students, track individuals, infer attendance, engagement, learning, teaching quality or productivity.

Room-level evidence

A room-level answer requires a correctly located sensor inside the room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

Authoritative-source precedence

Timetable, room, booking, capacity, access, accessibility, AV/IT, closure, maintenance, visitor and emergency status come from the authorised owner. Sensors cannot grant permission, bookability or safety approval.

Explicit operation type

Each record distinguishes direct observation, authorised lookup, calculation, estimate, forecast, comparison, diagnosis and recommendation. Answers should label which operation was performed.

Evidence quality and abstention

Use timestamps, aggregation, calibration/commissioning, uptime, gaps, drift, source agreement and provenance. Ask for clarification, narrow the answer or return not assessable when evidence is inadequate.

Privacy, accessibility and safety

Use anonymous aggregate occupancy and numerical acoustic indicators. Preserve permission boundaries and accessible alternatives. Certified alarms and official emergency instructions always take precedence.

Stakeholder value

The output should help a lecturer make a practical teaching decision, not create an operational dashboard or shift staff responsibilities onto the conversational service.

2. How to read each question record

Every record turns a natural teaching question into a bounded evidence requirement. The fields define minimum evidence, authoritative precedence, operation type, quality and uncertainty requirements, and when the service must qualify, defer or refuse.

LECTURER/TUTOR QUESTION	A natural question owned by teaching staff. It is phrased around a preparation, delivery, support, accessibility, contingency or evidence decision rather than an operator task.
SENSORS AND TELEMETRY	Physically plausible proposed room, route, AV, network, BMS, energy or equipment evidence. The field also states location, timing and quality conditions relevant to the question.
AUTHORITATIVE SUPPORTING DATA	Current timetable, room, capacity, access, booking, accessibility, AV/IT, visitor, maintenance, policy or safety records that must take precedence over sensor inference.
ANALYSIS REQUIRED	The exact operation: observation, lookup, calculation, estimate, forecast, comparison, diagnosis or recommendation. It identifies minimum evidence, time window and appropriate uncertainty.
ANSWER BOUNDARY	What the building must not infer, disclose, control or certify; when it must ask for clarification, provide a limited answer, defer to an authority or return not assessable.
WHY THIS IS USEFUL	The concrete teaching decision supported by the answer.

Readiness

R1 - practical

Core proposed sensing plus basic current spatial, service or provenance data.

R2 - integration-dependent

Timetable, room, booking, access, accessibility, AV/IT, BMS, maintenance, visitor or forecasting integration.

R3 - research-grade or restricted

Dense sensing, validated comparison/diagnosis, controlled operational data, or stronger privacy, governance and safety controls.

Global rule: use only verified eligible spaces and authorised data; preserve timestamps and room-level coverage; inspect no teaching, speech or network content; never infer identity, attendance or performance; and return a limited or not-assessable answer when critical evidence is missing.

3. Worked example: a demanding teaching day

A responsible answer combines fixed teaching commitments and lecturer-supplied requirements with verified room, access, AV, accessibility and environmental evidence. It never treats an empty room as available, a quiet room as private, or environmental telemetry as permission or safety certification.

Lecturer: "I have a 9 a.m. lecture, a hybrid seminar at noon, office hours at 2 p.m. and an evening guest talk. Can you identify the setup risks, accessible routes, verified fallback options and where the evidence is too weak to recommend?"

1. Resolve the day

Resolve current room assignments, fixed times, setup windows, route, AV, privacy, accessibility, visitor and evening constraints from the minimum necessary lecturer-supplied information.

2. Verify authoritative status

Use current timetable, room purpose, capacity, access, bookings, AV/IT, accessibility, visitor approval, opening, maintenance and closure records before consulting sensors.

3. Verify evidence quality

Require target-room environmental coverage, current device/service status, timestamps, calibration/commissioning, uptime, gaps and valid route telemetry. Mark unverified criteria unknown.

4. Perform the right operation

Use lookup for permissions/status, observation for current room conditions, calculation for routes, forecast for future risk and diagnosis only when time-aligned evidence supports it.

5. Rank lecturer-owned actions

Recommend arrival buffers, breaks, verified room requests, approved fallbacks and recheck points; do not book rooms, change controls or take over AV/IT, facilities or safety decisions.

6. Communicate answerability

State sources, observed/retrieved times, confidence, omitted criteria and safe failure. Return a conditional or not-assessable result when essential evidence or authority is missing.

ILLUSTRATIVE ANSWER STRUCTURE

"The 9 a.m. room is confirmed by the current timetable, but AV readiness is unknown, so arrive 15 minutes early and run the approved check. The noon hybrid room reports ready at [time]; network risk is [low/moderate] based on aggregate service telemetry, not traffic content. Office-hour option [A] is verified bookable and accessible; low occupancy alone was not used as privacy evidence. The evening talk remains conditional on visitor approval and after-hours access. Recheck lift and closure status at [time]."

1. Session and room preparation

Questions LT-001-LT-002

LT-001

CORE | L2 - readiness synthesis | R2 - integration-dependent

Lecturer/tutor question: I have a 9 a.m. lecture. Before I leave my office, what is confirmed, what is still unknown, and how early should I arrive?

Sensors and telemetry

Proposed evidence: target-room temperature, RH, direct NDIR CO_2 and illuminance; anonymous room occupancy; authorised status events for installed AV, room power, lift and access doors. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current personal timetable and room assignment; room purpose, capacity, staff access and opening hours; AV/IT readiness, accessibility, planned works, closures and support status. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: authorised lookup, current observation and readiness recommendation. Check assignment, access and service constraints first; then summarise room conditions. Require current assignment, access and AV/service status plus room evidence no older than 5 minutes; report unknowns and confidence in the arrival buffer.

Answer boundary

The service cannot guarantee every cable, seat or personal device, grant early access or certify the room as safe. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Supports the lecturer to target preparation time at verified risks and avoid an unnecessary or late room visit.

LT-002

CORE | L2 - comparison + recommendation | R2 - integration-dependent

Lecturer/tutor question: I have back-to-back seminars in different rooms. Which transition needs the larger travel and setup buffer?

Sensors and telemetry

Proposed evidence: target-room environmental status, anonymous room occupancy and doorway flow; authorised AV start-up, room-power and lift status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, assigned rooms, permitted setup windows, staff access, route graph, room features, AV configuration, known incidents and closures. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: lookup, route calculation, comparison and recommendation. Estimate travel, access and setup from verified room and equipment differences, supplemented by current service state and comparable historical start-up times. Report a setup-time range, source gaps and which difference drives the recommendation.

Answer boundary

Do not infer availability from low occupancy or inspect teaching materials, device content or network traffic. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Supports the lecturer to sequence travel and preparation realistically so the second seminar is less likely to start late.

1. Session and room preparation

Questions LT-003-LT-004

LT-003

CORE | L2 - timed readiness check | R2 - integration-dependent

Lecturer/tutor question: Can I receive one concise, source-timestamped readiness check 30 minutes before each class?

Sensors and telemetry

Proposed evidence: room-level temperature, RH, CO_2, illuminance and anonymous occupancy; authorised AV, network-service, room-power, door, lift and BMS fault/status events. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: personal timetable trigger, assigned room, permitted access window, current incidents, planned works, accessibility status, support availability and building opening information. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: scheduled authorised lookup plus current observation. At the agreed time, classify the minimum readiness set as confirmed, degraded, unknown or not applicable; state source timestamps and route unresolved service issues to the correct owner. Classify each required source as confirmed, degraded, unknown or not applicable; preserve source timestamps and do not infer a missing service state.

Answer boundary

The alert must not reveal another class, identify occupants, grant access or claim safety from normal telemetry. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the observation not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Provides one dependable pre-class summary without requiring the lecturer to check several systems.

LT-004

HIGH | L3 - forecast + comparison | R2 - integration-dependent

Lecturer/tutor question: The room felt stuffy last week. Given today's class size and conditions, is the same pattern likely to recur?

Sensors and telemetry

Proposed evidence: direct NDIR CO_2, temperature and RH inside the target room; anonymous occupancy; supply-airflow, fan and damper telemetry with verified room-to-system mapping; outdoor weather. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room assignment, session duration, expected class size, room capacity, historical timetable context, and current HVAC incidents or planned work. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: matched historical comparison and forecast. Compare the earlier session with periods similar in duration, occupancy and weather; forecast CO_2 and temperature trajectories, state uncertainty and give a recheck time before or during class. Report sample count, forecast interval, missingness, confounders and a pre-class recheck time.

Answer boundary

A forecast is not a guarantee, and CO_2 is a ventilation indicator rather than a complete air-quality or comfort measure. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Supports a timely decision about breaks, arrival checks or requesting ventilation support before disruption recurs.

1. Session and room preparation

Questions LT-005-LT-006

LT-005

CORE | L2 - contingency lookup | R2 - integration-dependent

Lecturer/tutor question: If my assigned room is withdrawn, which currently eligible alternative should I ask the room team to consider?

Sensors and telemetry

No sensor establishes room eligibility or bookability. Contextual telemetry, if used, must be current, permissioned and correctly mapped; it cannot substitute for the named authoritative record.

Authoritative supporting data

Authoritative data: current timetable and room-booking systems; teaching-room eligibility, capacity, staff access, accessibility, installed equipment, layout, closures and maintenance. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: authorised lookup and constrained recommendation. Apply time, capacity, teaching-use, access, accessibility and equipment constraints first; then rank verified candidates by route, current condition and service readiness, with an independent second option. Apparent vacancy is not evidence of availability.

Answer boundary

An apparently empty room is not necessarily bookable. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Provides the lecturer with a realistic contingency to discuss quickly with the authorised room or timetabling team.

LT-006

HIGH | L2 - setup-time estimate | R2 - integration-dependent

Lecturer/tutor question: I am bringing demonstration equipment. How much authorised setup time should I allow, including access, power and safe-placement checks?

Sensors and telemetry

Proposed evidence: authorised equipment status where instrumented, room temperature/RH, room-power state, lift and door status, and aggregate corridor footfall for route delay. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room assignment, equipment booking and ownership, permitted use, staff induction/access, delivery route, lift constraints, setup window, approved risk controls and support availability. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: authorised lookup, route calculation and bounded setup-time estimate. Add verified travel, access, equipment warm-up, connection and required safety-check durations; use congestion or service state only as transparent adjustments and show assumptions. Report the combined range, assumptions, unavailable inputs and an authorised fallback.

Answer boundary

The estimate cannot authorise equipment use or replace induction, risk assessment, manufacturer procedure or technical support. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the calculation or estimate not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Helps the demonstrator avoid rushed preparation while keeping equipment and safety decisions with the competent owners.

1. Session and room preparation

Questions LT-007-LT-008

LT-007

CORE | L2 - suitability check | R2 - integration-dependent

Lecturer/tutor question: I want to alternate between a short lecture and small-group discussion. Does the assigned room currently support that format without blocking accessible routes?

Sensors and telemetry

Proposed evidence: room-level anonymous occupancy, numerical sound level, temperature, CO_2 and illuminance may describe operating conditions. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room purpose, approved capacity, furniture/layout record, accessible seating and circulation, installed teaching equipment, booking conditions and relevant safety limits. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: authoritative lookup plus suitability calculation. Compare expected attendance and activity requirements with verified layout, equipment and circulation; add environmental constraints only where valid room-level observations exist. Use environmental data only as valid room-level context, not as proof of suitability.

Answer boundary

Do not infer layout from the May 2021 plan or occupancy, advise moving fixed or safety-critical furniture, or treat a quiet room as permission to reconfigure it. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Allows the lecturer to adapt the teaching plan before class rather than discovering that the room cannot support the activity.

LT-008

HIGH | L3 - conditional recommendation | R3 - research-grade or restricted

Lecturer/tutor question: If ventilation feels poor during class, which lecturer actions are actually permitted here, and what trade-offs should I know?

Sensors and telemetry

Proposed evidence: direct room CO_2, temperature, RH and PM2.5; verified window/door state; outdoor temperature, PM and weather; numerical indoor/outdoor sound levels; HVAC airflow/state. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; numerical acoustic or flicker metrics only; no audio, speech or images. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: whether the opening is operable and permitted; fire-door, security, accessibility and HVAC guidance; current weather and outdoor-air source; any local restriction or maintenance notice. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: authorised lookup, current observation and conditional recommendation. Apply permission and safety constraints first, then compare likely ventilation benefit against outdoor pollution, weather, noise and HVAC interaction; include a recheck condition. Compare current indoor/outdoor conditions and likely trade-offs; withhold a recommendation if any hard constraint is unknown.

Answer boundary

Do not advise opening or propping a fire/security door, override a building rule, or treat telemetry as ventilation or air-quality certification. If permission, current indoor/outdoor evidence or a safety constraint is unresolved, give no action recommendation and defer to the room or facilities owner.

Why this is useful: Supports a permitted, proportionate in-class response without bypassing fire, security, accessibility or building-control rules.

1. Session and room preparation

Questions LT-009-LT-010

LT-009

HIGH | L3 - session forecast | R2 - integration-dependent

Lecturer/tutor question: For this two-hour class, is there evidence to plan a short break, and when would a recheck be most useful?

Sensors and telemetry

Proposed evidence: distributed room CO_2, temperature and RH in large spaces; anonymous occupancy or doorway counts; supply airflow and outdoor weather. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current session duration, expected class size, assigned room and capacity, comparable previous sessions, present HVAC incidents and any timetable constraint on a break. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: forecast and teaching recommendation. Estimate when CO_2 or temperature is likely to reach a selected operational trigger under comparable loads; report the range and confidence, then suggest a break or recheck point rather than a fixed rule. Report persistence, prediction interval and a recheck point rather than treating one high reading as decisive.

Answer boundary

The service cannot infer individual comfort or health need, and no single threshold determines when every class needs a break. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Supports a planned pause before conditions affect the session instead of reacting only after students complain.

LT-010

CORE | L2 - evidence explanation | R1 - practical

Lecturer/tutor question: When you call a room 'ready for teaching', which checks are confirmed, which are inferred, and which remain unknown?

Sensors and telemetry

Use provenance from every cited environmental, occupancy, AV, power, network, lift and door source: device identity, spatial mapping, observed/retrieved time, aggregation, commissioning/calibration, health, uptime and missingness. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, room eligibility, access, capacity, accessibility, AV/IT status, maintenance, closures and safety information, including source owner and effective time. Current timetable, room, access and service-owner records, with effective/retrieved time and lecturer permission, take precedence over telemetry, apparent vacancy and the May 2021 plan.

Analysis required

Operation: evidence explanation. Separate authoritative lookups, direct observations, calculations, estimates, forecasts and recommendations; list the minimum readiness criteria, omitted criteria, timestamps, confidence and any unresolved conflict. For every conclusion, label source owner, location, observed/effective/retrieved time, operation type, quality, permission and unresolved conflict.

Answer boundary

Do not use fluent wording to hide an inference or missing source. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the room, timetable or teaching-support owner.

Why this is useful: Lets the lecturer understand exactly what a readiness label can and cannot support before relying on it.

2. Lecture, seminar, tutorial and practical delivery

Questions LT-011-LT-012

LT-011

CORE | L2 - live observation + recommendation | R2 - integration-dependent

Lecturer/tutor question: During this lecture, are conditions changing persistently enough for me to pause, take a break or contact support?

Sensors and telemetry

Proposed evidence: current room CO_2, temperature and RH, with distributed points in a large room; anonymous occupancy; verified supply-airflow or HVAC status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room and session window, room capacity, support contact, active HVAC incidents and any approved in-class response guidance. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: live observation and bounded recommendation. Test whether a change is sustained across valid room points, compare with the session baseline and distinguish an environmental trend from a confirmed fault. State whether the evidence shows a trend, not a confirmed mechanical fault.

Answer boundary

Do not diagnose a mechanical failure from environmental data alone or claim that conditions are unsafe. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Supports the lecturer to respond proportionately during class without reacting to every short-lived sensor fluctuation.

LT-012

HIGH | L3 - spatial comparison | R3 - research-grade or restricted

Lecturer/tutor question: If this large room has validated zone-level coverage, which seating area currently has the best evidenced balance of ventilation, temperature, light and audibility?

Sensors and telemetry

Proposed evidence: distributed room coverage with CO_2, temperature/RH, illuminance and numerical acoustic sensors across the occupied zones, plus supply-airflow context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; numerical acoustic or flicker metrics only; no audio, speech or images. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room layout, accessible seating, fixed teaching position, capacity, permitted seating areas and any temporary closure. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: spatial comparison and conditional recommendation. Compare zone-level observations over a recent stable window, preserve each criterion separately, apply accessibility and permitted-use constraints, and report trade-offs and confidence. Rank zones only when coverage is sufficient; report sensor disagreement, interpolation limits and uncertainty.

Answer boundary

Do not interpolate zones from one sensor, guarantee individual comfort or hearing, or direct students to reserved or inaccessible seating. Without simultaneous validated zone coverage and current seating/access data, return room-wide facts only and mark the zone recommendation not assessable.

Why this is useful: Supports a zone-level seating suggestion only when the room has genuine spatial coverage and no access constraint is breached.

2. Lecture, seminar, tutorial and practical delivery

Questions LT-013-LT-014

LT-013

CORE | L2 - sustained-condition alert | R2 - integration-dependent

Lecturer/tutor question: Can you notify me only when a verified room or service condition persists beyond an agreed teaching threshold?

Sensors and telemetry

Proposed evidence: room-level environmental sensing and authorised AV/IT/BMS service events, each with timestamps, quality state and persistence. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: lecturer-selected notification window, current session schedule, approved service thresholds or incident states, room mapping and support escalation route. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: sustained-condition detection and notification. Apply separate persistence, severity and multi-source rules for environmental and service signals; suppress duplicates, state the evidence window and identify whether the notice is observation or authorised incident lookup. Suppress isolated spikes and state the trigger, duration, evidence quality and authorised response route.

Answer boundary

The service must not invent safety thresholds, suppress certified alarms, infer student impact or replace an official incident channel. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the observation not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Reduces alert fatigue while keeping attention on conditions that may justify a teaching adjustment.

LT-014

HIGH | L2 - acoustic context estimate | R2 - integration-dependent

Lecturer/tutor question: Is corridor noise likely to interfere with this tutorial during the next hour, and how confident is that forecast?

Sensors and telemetry

Proposed evidence: numerical acoustic indicators inside the tutorial room and in the adjacent corridor, with no retained audio; corridor footfall and door state. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current tutorial room, nearby timetable or event schedule, room-door characteristics, permitted occupancy and any planned works. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: current comparison and short-horizon estimate. Use current room and corridor numerical acoustics, door state, nearby timetable and class-change flow. Relate recent corridor sound/footfall patterns and scheduled changeovers to the tutorial window; distinguish likely background disturbance from measured speech intelligibility and state uncertainty.

Answer boundary

Numerical sound data cannot identify speakers, conversation content or guarantee audibility for every participant. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Helps the tutor choose when to close the door, adjust activity timing or seek an alternative verified room.

2. Lecture, seminar, tutorial and practical delivery

Questions LT-015-LT-016

LT-015

HIGH | L3 - diagnostic reasoning | R3 - research-grade or restricted

Lecturer/tutor question: The practical room is heating up. Does the evidence favour occupancy, equipment load, solar gain or an HVAC issue?

Sensors and telemetry

Proposed evidence: target-room temperature/RH at representative points, anonymous occupancy, equipment/circuit power, solar radiation or facade context, supply/return temperature, airflow, fan and valve/damper state. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current practical schedule, expected activity, authorised equipment inventory/status, room orientation, HVAC incidents, maintenance and control overrides. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: time-aligned diagnostic ranking. Check onset, persistence and lags across occupancy, load, solar and HVAC variables; compare with matched sessions and rank supported explanations without converting association into proven cause. Test onset and lags against unaffected periods, rank competing explanations and avoid causal claims.

Answer boundary

The service must not inspect experiment or teaching content, operate equipment/HVAC or declare a fault without corroborating operational evidence. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the diagnosis not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Gives the demonstrator a defensible basis for an immediate teaching adjustment and a targeted support request.

LT-016

HIGH | L3 - recovery analysis | R2 - integration-dependent

Lecturer/tutor question: Did the previous class leave too little recovery time before mine, or are today's conditions within the room's normal turnaround pattern?

Sensors and telemetry

Proposed evidence: room CO_2, temperature and RH, anonymous occupancy/doorway counts and verified HVAC airflow/state. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: consecutive timetable entries, room assignment, scheduled changeover, active HVAC incidents and room capacity. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: changeover recovery calculation. Identify the previous session end, vacancy interval and current start; calculate whether conditions returned toward a matched unoccupied baseline and separate carry-over from a new rise after entry. Estimate recovery against the room baseline while controlling for occupancy, HVAC and weather; report gaps and uncertainty.

Answer boundary

Do not identify or blame the previous class, infer exact attendance or treat non-recovery as a confirmed HVAC fault. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the calculation or estimate not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Supports a lecturer decision about whether to ventilate, delay a high-load activity or report an inadequate changeover pattern.

2. Lecture, seminar, tutorial and practical delivery

Questions LT-017-LT-018

LT-017

MEDIUM | L3 - matched comparison | R3 - research-grade or restricted

Lecturer/tutor question: For my recurring seminar, does either eligible room or time slot give more stable conditions under comparable use?

Sensors and telemetry

Proposed evidence: comparable room-level CO_2, temperature/RH, illuminance and numerical sound data, anonymous occupancy and relevant HVAC status across both options. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: historical and current timetable, eligible room options, capacity, accessibility, equipment, closures and maintenance changes. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: matched longitudinal comparison and recommendation. Compare equivalent seminars by duration, occupancy band, weather and term period; assess stability and disruption frequency for each criterion, then present a transparent lecturer-selected ranking. Report effect size, sample count, configuration changes and uncertainty for each option.

Answer boundary

Do not use student identities, attendance records or teaching outcomes, and do not imply future availability. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Provides evidence for a future room or time request based on repeat teaching conditions rather than one difficult session.

LT-018

CORE | L2 - session comparison | R2 - integration-dependent

Lecturer/tutor question: How do today's room conditions compare with my earlier sessions here, using only aggregate, non-identifying evidence?

Sensors and telemetry

Proposed evidence: target-room environment, anonymous occupancy bands and authorised aggregate service events for the current and prior session windows. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: lecturer's historical session times and rooms, current assignment, room/service changes and relevant maintenance or closure records. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: matched session comparison. Compare environmental levels, variability and confirmed service disruptions after adjusting for duration, occupancy band and weather where available; show differences, sample counts and uncertainty. Report differences, completeness and uncertainty without individual attendance data.

Answer boundary

Do not infer attendance, engagement, learning or lecturer performance. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Shows whether today was genuinely unusual and helps the lecturer decide whether a recurring issue needs escalation.

2. Lecture, seminar, tutorial and practical delivery

Questions LT-019-LT-020

LT-019

HIGH | L2 - task-lighting assessment | R2 - integration-dependent

Lecturer/tutor question: Are the desk, display and board areas adequately lit for this session, and where is the evidence incomplete?

Sensors and telemetry

Proposed evidence: illuminance at representative desk and presentation zones, daylight/solar context and authorised lighting state; optional calibrated flicker measurement where justified. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room layout, display/board positions, approved teaching use, lighting controls, accessibility requirements and any maintenance incident. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: task-specific observation and suitability assessment. Compare representative zones over the teaching window, identify glare or unevenness indicators, and state which task or location each observation supports. Distinguish measured illuminance from estimated glare or flicker and report uncovered areas.

Answer boundary

Do not claim visual comfort for every student, infer disability or alter lighting controls. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, facilities or teaching-support owner.

Why this is useful: Supports the lecturer to adjust room setup or request support before poor visibility disrupts the session.

LT-020

HIGH | L3 - post-activity recovery forecast | R3 - research-grade or restricted

Lecturer/tutor question: After this practical demonstration, when may the next group use the room, and which part of that answer is a forecast rather than formal release?

Sensors and telemetry

Proposed evidence: room temperature/RH, CO2, PM2.5/PM10 and indicative TVOC where the activity justifies them; extraction/HVAC airflow, equipment power and anonymous occupancy. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current and next bookings, approved activity/risk controls, required shutdown or clearance procedure, equipment status, extraction guidance and room owner release process. Current session, room and authorised BMS or service-owner records, with effective/retrieved time and lecturer permission, take precedence over sensor interpretation.

Analysis required

Operation: recovery observation and forecast. Detect the post-activity peak, compare decay with the verified pre-activity baseline and authoritative clearance criteria, estimate a recheck time and separate environmental recovery from formal room release. Use the verified pre-activity baseline, post-activity peak, extraction/HVAC state and approved clearance criteria.

Answer boundary

Indicative TVOC or particle sensing cannot certify that an activity is safe, and telemetry cannot release the room. If approved clearance criteria, extraction state, room evidence or formal owner release is missing, report only the measured trend and defer use of the room to the responsible owner.

Why this is useful: Separates environmental recovery from formal room release so the lecturer does not treat a forecast as permission.

3. Room suitability, accessibility, comfort and student support

Questions LT-021-LT-022

LT-021

CORE | L2 - accessible-route recommendation | R2 - integration-dependent

Lecturer/tutor question: A student needs a step-free route and wants to avoid the busiest corridor. What current route and arrival buffer can I suggest?

Sensors and telemetry

Proposed evidence: aggregate corridor/stair footfall and doorway counts; authorised live lift and door status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current accessible-route graph, entrance and room access, lift status, temporary closures, teaching time and any user-supplied mobility constraint. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: authorised route lookup, time-dependent path calculation and recommendation. Exclude non-step-free or closed segments, estimate class-change congestion, and provide a primary route, travel buffer and verified fallback. Provide a travel-time range, primary route and verified fallback; expose uncovered segments.

Answer boundary

Do not infer disability, guarantee crowd levels or present the route as accessibility certification. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the room or accessibility owner.

Why this is useful: Supports the lecturer to give practical, current guidance while respecting individual choice and authoritative access information.

LT-022

CORE | L2 - private-space recommendation | R2 - integration-dependent

Lecturer/tutor question: Which currently authorised space could I request for an office-hour conversation needing step-free access and reasonable acoustic privacy?

Sensors and telemetry

Proposed evidence: anonymous occupancy, numerical sound level and current room environment may help rank already eligible spaces. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room-booking and staff-access systems, permitted use, capacity, step-free route, door/room privacy features, opening hours and current closures. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: authorised lookup and constrained recommendation. Filter by bookability, access, permitted use and accessibility; then compare route, current occupancy and numerical noise, clearly separating quietness from verified privacy. Low sound or low occupancy does not establish confidentiality.

Answer boundary

Do not infer confidentiality from low sound or low occupancy, reveal who is using a room or reserve it. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room or accessibility owner.

Why this is useful: Helps the tutor arrange a suitable student conversation without compromising privacy, access or room ownership.

3. Room suitability, accessibility, comfort and student support

Questions LT-023-LT-024

LT-023

HIGH | L3 - inclusive-room recommendation | R3 - research-grade or restricted

Lecturer/tutor question: For a seminar where some students benefit from lower noise and stable lighting, which eligible room is the strongest evidenced option?

Sensors and telemetry

Proposed evidence: room-level numerical acoustic, illuminance, optional validated flicker, temperature and CO2 data with equivalent placement and time coverage; anonymous occupancy for context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current eligible room options, capacity, accessibility, assistive systems, layout, permitted use, bookings, maintenance and any user-stated teaching requirements. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: constrained multi-criteria comparison and recommendation. Apply eligibility, capacity and accessibility first; compare stable teaching-period evidence for noise and lighting; present each criterion, uncertainty and a second option. Compare compatible room-level noise, lighting and environmental windows, showing criterion weights, evidence gaps and a second option.

Answer boundary

Do not label a room universally accessible, guarantee sensory comfort or disclose student information. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room or accessibility owner.

Why this is useful: Supports an inclusive room request grounded in current features and environmental evidence rather than assumptions.

LT-024

CORE | L2 - capacity and circulation check | R2 - integration-dependent

Lecturer/tutor question: Is the assigned room currently suitable for the expected class size, including verified capacity, accessible seating and circulation?

Sensors and telemetry

Proposed evidence: anonymous occupancy and doorway counts can describe current load but cannot verify approved capacity, furniture, accessible seating or egress. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current approved capacity, room layout, accessible seating/circulation, teaching purpose, booking, temporary restrictions and relevant safety information. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: authorised lookup and capacity/circulation check. Compare expected class size with the current approved configuration; report confirmed constraints and use current occupancy only to describe immediate crowding, not formal suitability. Treat anonymous occupancy only as current load context; do not infer formal suitability from a count.

Answer boundary

Do not infer capacity or safety from sensor counts, identify attendees or advise exceeding an authorised limit. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room or accessibility owner.

Why this is useful: Allows the lecturer to raise a room-fit problem before students arrive and avoid an inaccessible or over-constrained setup.

3. Room suitability, accessibility, comfort and student support

Questions LT-025-LT-026

LT-025

HIGH | L2 - group-work suitability | R2 - integration-dependent

Lecturer/tutor question: Can this room support the planned small-group layout without obstructing accessible routes or exceeding usable capacity?

Sensors and telemetry

Proposed evidence: anonymous occupancy and zone-level presence, room CO_2/temperature and numerical sound may describe expected operating conditions. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room layout, movable/fixed furniture, approved capacity, accessible circulation, group-work permission, expected class size and booking conditions. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: suitability calculation and conditional recommendation. Test the proposed group arrangement against verified layout and circulation, then estimate crowding, acoustic and ventilation implications from comparable activity where evidence exists. Estimate crowding, ventilation and noise only from valid room evidence and expected activity; state assumptions.

Answer boundary

Do not direct furniture moves that conflict with room rules or safety, infer individual comfort or treat occupancy estimates as exact. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the room or accessibility owner.

Why this is useful: Helps the tutor choose a workable activity format before rearrangement creates access or crowding problems.

LT-026

HIGH | L2 - evidence-limited environmental response | R2 - integration-dependent

Lecturer/tutor question: A student reports headache and stuffiness. What environmental facts can I relay, and which health conclusions must the service avoid?

Sensors and telemetry

Proposed evidence: current room CO_2, temperature, RH, PM2.5 and indicative TVOC where installed, plus ventilation/HVAC status and sensor-health metadata. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room/session, active maintenance or ventilation incidents, approved support and emergency guidance, and any lecturer-supplied observation. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: current observation and evidence-limited response. Separate environmental context from health advice; do not diagnose the student or delay official urgent-help procedures. Report only verified room observations, recent trends and authorised incident status.

Answer boundary

Do not diagnose the cause of a headache, give medical advice or identify the student. Report only quality-controlled room observations and official support routes; urgent symptoms or suspected danger must follow current first-aid or emergency procedures.

Why this is useful: Provides useful room facts while avoiding unsupported medical explanation or advice.

3. Room suitability, accessibility, comfort and student support

Questions LT-027-LT-028

LT-027

MEDIUM | L2 - waiting-area recommendation | R2 - integration-dependent

Lecturer/tutor question: Where can students wait before office hours without obstructing circulation or an accessible route?

Sensors and telemetry

Proposed evidence: anonymous occupancy and aggregate footfall in public waiting/collaboration areas, with numerical noise and room environment where relevant. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current public waiting areas, access and permitted use, accessible-route clearances, opening hours, closures and office-hour location. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: authorised lookup and short-horizon recommendation. Exclude restricted or circulation-critical locations, compare current and expected footfall, route distance and capacity, and give a primary and fallback waiting area. Rank by route, capacity, aggregate footfall and numerical noise over the next office-hour window, with a fallback and no disclosure of who is present.

Answer boundary

Do not direct students into staff-only or bookable rooms, disclose who is waiting or guarantee future space. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room or accessibility owner.

Why this is useful: Reduces corridor obstruction and gives students a clearer, more inclusive office-hour experience.

LT-028

MEDIUM | L3 - matched room comparison | R3 - research-grade or restricted

Lecturer/tutor question: Is this room persistently colder than comparable teaching rooms, or is today's episode unusual?

Sensors and telemetry

Proposed evidence: room temperature/RH and, for stronger comfort evidence, radiant temperature and airflow; verified HVAC state and outdoor weather. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room identity and HVAC zone, comparable teaching-room set, timetable, occupancy band, control changes and incidents. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: matched longitudinal comparison. Compare the room with similar rooms and its own history under similar weather, occupancy and schedule; quantify frequency, duration and magnitude of cold episodes with uncertainty. Quantify frequency, duration and magnitude of cold episodes with confidence intervals.

Answer boundary

Do not claim individual discomfort, diagnose a fault solely from temperature or compare rooms with incompatible sensing. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room or accessibility owner.

Why this is useful: Supports a lecturer decision about whether to report a persistent room problem or treat the experience as an isolated event.

3. Room suitability, accessibility, comfort and student support

Questions LT-029-LT-030

LT-029

MEDIUM | L3 - visual-comfort assessment | R3 - research-grade or restricted

Lecturer/tutor question: Is there enough task light for note-taking without likely glare on the display or board?

Sensors and telemetry

Proposed evidence: illuminance at representative desk and presentation zones, daylight/solar context and authorised lighting state; optional luminance/glare measurements where a validated method is available. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room layout, display/board locations, window/blind configuration, approved teaching use, accessibility requirements and lighting maintenance. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: visual-task observation and assessment. Compare desk illuminance and presentation-zone contrast over the session period, identify unevenness or likely glare conditions and state which criterion is directly measured versus estimated. Report measured illuminance separately from any validated glare or flicker estimate.

Answer boundary

Do not guarantee comfort for every viewer, infer visual impairment or operate blinds/lights without authority. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room or accessibility owner.

Why this is useful: Supports the lecturer to adjust presentation choices or request support before visibility affects participation.

LT-030

CORE | L2 - alternative-location guidance | R2 - integration-dependent

Lecturer/tutor question: The usual step-free route is unavailable. Which verified alternative route or room option should I ask the responsible team about?

Sensors and telemetry

No sensor determines room eligibility. Contextual telemetry, if used, must be current, permissioned and correctly mapped; it cannot substitute for the named authoritative record.

Authoritative supporting data

Authoritative data: current room assignment, accessible routes, lift/door outages, room eligibility, capacity, booking, access, required teaching features and temporary closures. Current room, access and accessibility records and competent advice, with effective/retrieved time and permission, take precedence over occupancy or environmental inference.

Analysis required

Operation: authorised lookup, accessible-route calculation and constrained recommendation. Filter candidate spaces by current eligibility and teaching requirements, then rank by verified step-free travel and service readiness. Rank only authorised candidates by route and required teaching features; do not infer current use from the 2021 plan.

Answer boundary

The service cannot move the class, grant access or certify accessibility. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the room or accessibility owner.

Why this is useful: Provides the lecturer with a concrete, authorised contingency to raise when an accessibility barrier affects the scheduled session.

4. Audiovisual, hybrid and recorded teaching readiness

Questions LT-031-LT-032

LT-031

CORE | L2 - technical readiness synthesis | R2 - integration-dependent

Lecturer/tutor question: Before my hybrid seminar, which installed AV, network and power services are confirmed ready, degraded or unknown?

Sensors and telemetry

Authorised status and health for installed AV components, room power and aggregate network service; proposed room temperature where equipment overheating is relevant. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room assignment, installed AV inventory, approved hybrid configuration, service dashboard, incident/maintenance status, support availability and accessibility features. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: authorised lookup and technical-readiness synthesis. Check each required component against the verified room inventory and current health state, label confirmed/degraded/unknown, and provide the approved pre-session test or support route. Report each required component as ready, degraded or unknown with last update and maintenance state.

Answer boundary

A ready status cannot guarantee a personal device, cable, platform account or external participant connection. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Lets the seminar leader concentrate a short pre-class test on the components that actually need attention.

LT-032

MEDIUM | L3 - service reliability comparison | R3 - research-grade or restricted

Lecturer/tutor question: For future hybrid seminars, which currently eligible room has the strongest reliability evidence for the setup I need?

Sensors and telemetry

Authorised aggregate AV, room-computer, power and network service events; proposed room environmental context where heat-related degradation is relevant. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current eligible room set, capacity, accessibility, hybrid equipment, booking rules, maintenance changes and approved incident history. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: matched reliability comparison and recommendation. Compare verified incident rate, recovery time and readiness across equivalent session windows, account for equipment changes, and present each service dimension plus uncertainty. Account for equipment or software changes; report failure rate, recovery time, exposure and uncertainty.

Answer boundary

Do not inspect teaching or network content, attribute incidents to lecturers or students, or imply future room availability. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Provides evidence for a future hybrid-room request based on service performance rather than anecdote.

4. Audiovisual, hybrid and recorded teaching readiness

Questions LT-033-LT-034

LT-033

HIGH | L3 - cross-domain diagnosis | R3 - research-grade or restricted

Lecturer/tutor question: Could repeated AV failures be associated with room heat, equipment load or power events, or is the evidence insufficient to diagnose?

Sensors and telemetry

Authorised AV fault events, room and equipment temperatures, circuit or PDU power, UPS state where relevant, and HVAC/cooling status. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current AV inventory, incident timestamps, maintenance history, room assignment, planned works and authorised network/service status. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: cross-domain diagnostic ranking. Align AV failures with temperature, load, power and cooling events; test recurrence and lags, compare unaffected sessions, and rank supported explanations without claiming causality from coincidence. Compare unaffected sessions, test lags and rank supported hypotheses without inspecting content or claiming causation.

Answer boundary

Do not inspect device content, claim causality from timing alone or diagnose AV hardware from room telemetry. If service logs, asset mapping, power/environment data or permission are inadequate, describe the association and defer diagnosis to AV/IT support.

Why this is useful: Helps the lecturer give support teams a bounded, evidence-based fault pattern without pretending to diagnose equipment.

LT-034

CORE | L2 - privacy-preserving service alert | R2 - integration-dependent

Lecturer/tutor question: During a hybrid class, can you alert me to service degradation without inspecting the session, recording or network content?

Sensors and telemetry

Authorised aggregate health metrics for installed AV devices, room computer, power and network service, such as link state, packet-loss aggregate or device error status. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room/session, installed service inventory, support policy, incident state and lecturer-selected alert threshold. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: live service-state observation and sustained degradation alert. Detect persistent change in approved health metrics, suppress brief transients, state affected service, source time and confidence, and give the authorised fallback/support action. Identify the affected service, event time, confidence and approved fallback without analysing audio, video, screens or traffic content.

Answer boundary

Do not infer teaching quality, participant behaviour or content, and do not suppress official incident notices. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the observation not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Allows the lecturer to switch to an approved fallback before a hidden technical problem significantly disrupts remote participants.

4. Audiovisual, hybrid and recorded teaching readiness

Questions LT-035-LT-036

LT-035

CORE | L2 - capture readiness lookup | R2 - integration-dependent

Lecturer/tutor question: Is lecture capture scheduled and permitted for this session, and which installed components are confirmed ready?

Sensors and telemetry

Authorised lecture-capture scheduling and hardware health, room-computer/power/network status and device clock; no recording content, preview audio/video or participant data is accessed. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, capture booking/consent configuration, room inventory, institutional recording policy, access permissions, incident status and support contact. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: authorised lookup and readiness check. Confirm whether capture is scheduled, which approved components are required, their current health and any unresolved policy or service condition. Confirm current capture scheduling, permissions and required hardware against the room inventory, then check service health and maintenance state.

Answer boundary

The service cannot start, stop, publish or guarantee a recording, infer consent or inspect captured content. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Prevents the lecturer from assuming that a room booking automatically includes a working and authorised recording.

LT-036

HIGH | L2 - policy-based fallback | R2 - integration-dependent

Lecturer/tutor question: If authorised lecture capture is unavailable, which approved fallback applies to this session?

Sensors and telemetry

No environmental sensor is required. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current capture incident, institutional recording and accessibility policy, approved platforms, room equipment, support availability, permissions and any student-facing communication requirement. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: policy lookup and constrained recommendation. Apply consent, accessibility, security and service constraints, then present approved fallback options in priority order with setup steps and limitations. Rank only authorised fallbacks, state setup requirements and withhold advice if permission or accessibility is unresolved.

Answer boundary

Do not suggest unapproved personal recording, bypass consent/accessibility rules, inspect materials or create/publish content. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Provides the lecturer with a compliant alternative quickly when the normal capture service fails.

4. Audiovisual, hybrid and recorded teaching readiness

Questions LT-037-LT-038

LT-037

MEDIUM | L3 - congestion forecast | R3 - research-grade or restricted

Lecturer/tutor question: Is wireless presentation at elevated risk during the class-change peak, and what approved fallback should I prepare?

Sensors and telemetry

Authorised aggregate wireless-service health for the relevant room/area, such as access-point availability, utilisation band and error rate; room device state and power. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room and session time, installed presentation system, planned network work, known incidents and historical aggregate service windows. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: matched historical comparison and short-horizon forecast. Compare equivalent class-change periods, account for maintenance/configuration changes, estimate degradation risk and state a wired or local approved fallback. Report a short-horizon probability range and an approved wired/offline fallback.

Answer boundary

The forecast cannot guarantee an individual device or account and must not infer attendance from network load. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Supports the lecturer to prepare the approved fallback before a predictable peak rather than losing teaching time.

LT-038

CORE | L2 - evening service check | R2 - integration-dependent

Lecturer/tutor question: Will my evening hybrid session be affected by closure, planned network work, access limits or reduced support?

Sensors and telemetry

Authorised live status for installed AV, power, network, lift and access systems; proposed room environment only for present conditions. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, approved after-hours access, building opening/closure, planned works, network maintenance, support hours, room booking and lone-working or evening-use rules. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: authorised constraint lookup and readiness recommendation. Check hard access and service conditions, identify time-limited support or maintenance windows, and provide a verified fallback or recheck time where available. Report confirmed constraints and unknowns; telemetry cannot override closure or permission.

Answer boundary

Do not grant after-hours access, override closure or lone-working rules, or treat normal telemetry as permission. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Prevents an evening session from being planned around services or access that will not actually be available.

4. Audiovisual, hybrid and recorded teaching readiness

Questions LT-039-LT-040

LT-039

HIGH | L2 - assistive-listening readiness | R2 - integration-dependent

Lecturer/tutor question: Is the installed assistive-listening system authorised for this session and reporting ready?

Sensors and telemetry

Authorised health/status of the installed assistive-listening equipment, room power and relevant AV interface; optional numerical acoustic background indicators. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection; numerical acoustic or flicker metrics only; no audio, speech or images. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room inventory, assistive-system type, booking/loan requirements, accessibility guidance, maintenance state, support contact and session assignment. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: authorised lookup and readiness check. Confirm the verified room feature and current health, identify any required receiver or setup action, and distinguish equipment readiness from individual suitability. Verify the current assistive-listening inventory, latest authorised test, maintenance state and room configuration.

Answer boundary

Do not claim that the system meets every user need, infer a disability or substitute numerical sound levels for accessibility advice. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Supports the lecturer to confirm a key inclusive-teaching provision before students arrive.

LT-040

MEDIUM | L2 - post-session technical summary | R2 - integration-dependent

Lecturer/tutor question: After class, can you summarise confirmed service incidents and unresolved evidence without assessing attendance or teaching quality?

Sensors and telemetry

Authorised aggregate AV, room-computer, network, power and capture event/status logs with source timestamps; room environmental context only where technically relevant. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current session time/room, confirmed incident and maintenance records, service-owner classifications and resolution state. Current AV, capture and network service-owner records and permissions, with inventory version, event time and maintenance state, take precedence over device telemetry.

Analysis required

Operation: event matching and factual summary. Align confirmed service events with the session window, distinguish brief degradation from recorded incidents, show duration/source and identify unresolved evidence gaps. Report duration, source, resolution and evidence gaps without evaluating people or teaching.

Answer boundary

Do not infer attendance, engagement, learning, lecturer performance or blame; do not convert an unverified complaint into a confirmed fault. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the AV/IT or capture-service owner.

Why this is useful: Creates a factual post-session record for support follow-up without evaluating people or teaching.

5. Room transitions, timetable pressure and office-hour activity

Questions LT-041-LT-042

LT-041

CORE | L2 - time-dependent route calculation | R2 - integration-dependent

Lecturer/tutor question: I have ten minutes between rooms on different floors. Is the authorised route realistically achievable at class-change time?

Sensors and telemetry

Proposed evidence: aggregate corridor, stair and doorway footfall; authorised lift and door status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable and room assignments, staff access, accessible route graph, temporary closures, lift status and permitted setup times. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: time-dependent route calculation and estimate. Compute feasible routes, account for expected class-change congestion and lift state, include setup entry constraints and give a travel range plus fallback. Report a travel-time range, setup constraint and fallback route.

Answer boundary

Do not guarantee walking time, infer personal mobility or treat low occupancy as access permission. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the calculation or estimate not assessable and defer to the timetable, room or access owner.

Why this is useful: Supports a lecturer decision about whether to finish promptly, prepare material in advance or seek timetable support.

LT-042

MEDIUM | L3 - flow forecast + recommendation | R3 - research-grade or restricted

Lecturer/tutor question: Would ending this session a few minutes early materially reduce route congestion for my group, or is the evidence too uncertain?

Sensors and telemetry

Proposed evidence: aggregate doorway and corridor footfall with verified route coverage; no individual tracking. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, room exits, corridor capacity/route information, nearby session schedule and any teaching-time policy or accessibility consideration. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: historical flow comparison and bounded scenario estimate. Model group release at several small offsets, estimate overlap with nearby flows, and report relative change and uncertainty rather than a precise crowd count. Report expected congestion change, uncertainty and possible effects on timetable obligations.

Answer boundary

The service cannot alter teaching time, identify groups or certify circulation safety. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the timetable, room or access owner.

Why this is useful: Supports a small lecturer-owned timing adjustment that may improve movement without transferring corridor management to the lecturer.

5. Room transitions, timetable pressure and office-hour activity

Questions LT-043-LT-044

LT-043

HIGH | L2 - changeover readiness estimate | R2 - integration-dependent

Lecturer/tutor question: The previous class may overrun. When is the room officially released, and how much verified setup time will remain?

Sensors and telemetry

Proposed evidence: anonymous room occupancy and doorway flow; authorised room-computer/AV reset and cleaning/service status where available; room environment for recovery context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: consecutive timetable entries, permitted changeover/setup window, staff access, room-booking status and any reported overrun or service incident. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: authorised lookup, current observation and changeover-time estimate. Separate scheduled availability, observed vacancy and technical reset/recovery; state the earliest confirmed setup time and a recheck point. Use occupancy, door and cleaning/maintenance status only to estimate practical readiness; report a range and contact route when signals disagree.

Answer boundary

Vacancy does not grant entry or prove the class has officially ended. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the calculation or estimate not assessable and defer to the timetable, room or access owner.

Why this is useful: Reduces corridor waiting and rushed setup while respecting the previous class and room-access rules.

LT-044

CORE | L2 - office-hour recommendation | R2 - integration-dependent

Lecturer/tutor question: Between two classes, which authorised office-hour space adds the least travel while meeting privacy and access needs?

Sensors and telemetry

Proposed evidence: anonymous occupancy and numerical sound level may help rank already eligible spaces; route footfall and lift status may inform travel. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, available authorised rooms, permitted office-hour use, booking, staff access, step-free routes, opening hours and room privacy features. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: authorised lookup, route calculation and constrained recommendation. Filter by time, access, permitted use and accessibility; then compare detour, current occupancy and noise, clearly distinguishing quietness from privacy. Rank by route, aggregate footfall and numerical noise for the available interval, with a verified fallback.

Answer boundary

Do not direct the lecturer into an unbooked or restricted room, reveal occupants or describe a quiet open area as confidential. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the timetable, room or access owner.

Why this is useful: Supports the lecturer to provide student support without missing the next class or compromising privacy.

5. Room transitions, timetable pressure and office-hour activity

Questions LT-045-LT-046

LT-045

CORE | L1 - authorised notification | R2 - integration-dependent

Lecturer/tutor question: At what time am I officially allowed to enter this room and begin setup?

Sensors and telemetry

No sensor determines permission. Contextual telemetry, if used, must be current, permissioned and correctly mapped; it cannot substitute for the named authoritative record.

Authoritative supporting data

Authoritative data: current timetable and room booking, staff access policy, setup/changeover entitlement, door-control schedule, exceptional restrictions and room-owner guidance. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: authorised lookup. Use only current booking, room-owner and access-control records with effective times. Report the permitted setup window and any conflict; no sensor or apparent vacancy can grant permission.

Answer boundary

The service cannot grant access, override the previous booking or treat an empty room as available. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the timetable, room or access owner.

Why this is useful: Provides a clear answer to a common preparation question without relying on corridor observation or guesswork.

LT-046

HIGH | L3 - carry-over analysis | R2 - integration-dependent

Lecturer/tutor question: Are back-to-back sessions causing CO_2 or temperature to carry over into my class?

Sensors and telemetry

Proposed evidence: continuous target-room CO_2, temperature/RH, anonymous occupancy/doorway counts and verified HVAC airflow/state across multiple consecutive sessions. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: historical timetable, room capacity, session durations, maintenance/control changes and outdoor weather. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: longitudinal carry-over calculation. Estimate recovery during each changeover, compare with longer vacancy periods and matched weather/occupancy, and quantify recurrence and uncertainty. Compare with longer gaps under matched weather, occupancy and HVAC state; quantify recurrence and uncertainty.

Answer boundary

Do not identify classes, infer exact attendance or claim causation from sequence alone. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the timetable, room or access owner.

Why this is useful: Provides evidence for requesting a more realistic changeover or ventilation review for recurring teaching blocks.

5. Room transitions, timetable pressure and office-hour activity

Questions LT-047-LT-048

LT-047

HIGH | L2 - equipment-route planning | R2 - integration-dependent

Lecturer/tutor question: A visiting demonstrator is bringing equipment on a trolley. Which authorised route, entrance and arrival window should I give them?

Sensors and telemetry

Authorised live lift and door status; proposed aggregate corridor footfall for congestion context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: visitor approval, loading/entry arrangements, equipment dimensions and handling requirements, accessible/service route, lift capacity, staff escort/access, room booking and temporary closures. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: authorised route lookup, constraint calculation and timing recommendation. Exclude prohibited or unsuitable paths, estimate travel around class-change peaks, and provide a primary route, buffer and verified fallback. Calculate a travel and setup buffer, expose route gaps and provide an approved alternative.

Answer boundary

Do not grant visitor or service access, infer lift suitability from normal operation or bypass security and risk controls. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the timetable, room or access owner.

Why this is useful: Supports the lecturer to give the visitor practical instructions that reduce delay and avoid an inaccessible or unauthorised route.

LT-048

CORE | L2 - room-change comparison | R2 - integration-dependent

Lecturer/tutor question: My tutorial has moved rooms. Which verified differences in layout, accessibility, AV and environment should I prepare for?

Sensors and telemetry

Proposed evidence: current environment and anonymous occupancy may describe conditions; authorised AV, power, network, lift and door status may identify live constraints. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current replacement-room assignment, capacity, layout, accessibility, equipment, access, booking, route, opening and known incidents compared with the original room. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: authoritative room-feature comparison and readiness recommendation. Identify differences that affect the tutorial plan, travel, setup, accessibility and approved technology; add current condition only where room-level evidence is valid. Add current environmental/service context only where valid.

Answer boundary

Do not infer room features from the historical plan, assume equipment is usable or make a booking change. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the timetable, room or access owner.

Why this is useful: Lets the tutor adapt materials, activity and arrival time to the actual replacement room rather than discovering differences at the door.

5. Room transitions, timetable pressure and office-hour activity

Questions LT-049-LT-050

LT-049

MEDIUM | L2 - vacancy observation with authority | R2 - integration-dependent

Lecturer/tutor question: The room looks empty. Has it actually been released to my session, and are setup services available?

Sensors and telemetry

Proposed evidence: anonymous room occupancy and doorway flow can indicate likely vacancy; authorised door/access state may provide context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current booking and changeover window, previous-session release or room-owner status, staff access and any cleaning/maintenance hold. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: current observation plus authorised permission lookup. Report occupancy estimate separately from the official setup status and give a recheck or contact route if they disagree. Report anonymous vacancy evidence separately, with count uncertainty and timestamp; disagreement requires a recheck or room-owner contact.

Answer boundary

Do not identify occupants, announce that a class has ended from sensor data or grant entry from apparent vacancy. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the calculation or estimate not assessable and defer to the timetable, room or access owner.

Why this is useful: Prevents an apparently empty room from being mistaken for a room that is released, accessible and ready for setup.

LT-050

MEDIUM | L3 - office-hour forecast | R2 - integration-dependent

Lecturer/tutor question: When could I offer a short office-hour slot with lower corridor noise and foot traffic without clashing with my teaching?

Sensors and telemetry

Proposed evidence: aggregate corridor footfall and numerical sound indicators near eligible office-hour locations, with no audio or identity tracking. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: lecturer timetable, office-hour constraints, authorised spaces, booking/access, opening hours, accessible routes and nearby event/class schedule. Current timetable, room-release, access and route-service records, with effective/retrieved time and lecturer permission, take precedence over apparent vacancy or historical patterns.

Analysis required

Operation: schedule lookup, historical pattern comparison and recommendation. Find feasible time windows, forecast relative corridor activity and rank verified options by detour, accessibility and privacy features. Forecast the requested slot with an interval, exclusions and a second option; never guarantee quietness.

Answer boundary

Do not infer who is present, guarantee quietness or treat low traffic as privacy. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the timetable, room or access owner.

Why this is useful: Supports the lecturer to choose an office-hour time that is easier to reach and less disruptive for conversation.

6. Visiting speakers, events and irregular teaching

Questions LT-051-LT-052

LT-051

CORE | L2 - visitor-event readiness | R2 - integration-dependent

Lecturer/tutor question: A visiting speaker is coming. Which route, room, AV, access and environmental checks are confirmed before they arrive?

Sensors and telemetry

Proposed evidence: target-room environment and anonymous occupancy; authorised AV, room-computer, network, power, lift and door status; route footfall for congestion context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: visitor approval, seminar booking, room capacity/accessibility, arrival instructions, AV requirements, opening hours, support, planned works and closures. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: authorised lookup, current observation and event-readiness recommendation. Check visitor/access and room eligibility first, then service and environment; list confirmed, degraded and unknown items plus an arrival/setup buffer. Produce a timed checklist with unknowns, fallback and responsible contact.

Answer boundary

The service cannot approve the visitor, grant access, guarantee equipment or certify safety. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Gives the host lecturer one practical readiness view and focuses pre-event checks on unresolved items.

LT-052

MEDIUM | L2 - visitor waiting recommendation | R2 - integration-dependent

Lecturer/tutor question: Where may the visiting speaker wait without entering a restricted area or obstructing a busy corridor?

Sensors and telemetry

Proposed evidence: anonymous occupancy and aggregate footfall in public waiting areas, with numerical sound and room environment where relevant. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current public/visitor-permitted spaces, visitor route, host meeting point, opening hours, accessibility, closures and room booking. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: authorised lookup and short-horizon recommendation. Exclude restricted and circulation-critical areas, compare route distance, current occupancy and expected footfall, and provide a primary and fallback location. Do not disclose current users.

Answer boundary

Do not direct the visitor into staff-only or bookable space, reveal occupants or guarantee future availability. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Improves the visitor experience and reduces last-minute corridor congestion before the seminar.

6. Visiting speakers, events and irregular teaching

Questions LT-053-LT-054

LT-053

HIGH | L3 - arrival-flow forecast | R3 - research-grade or restricted

Lecturer/tutor question: For this public teaching event, where and when are queues most likely to form?

Sensors and telemetry

Proposed evidence: aggregate entrance, doorway and corridor counts with validated route coverage; room occupancy bands and lift status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: event schedule, verified room and entrance, expected registrations, access rules, route graph, temporary barriers, accessibility plan and nearby timetable. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: arrival-flow forecast and bottleneck comparison. Combine expected arrival waves with matched historical footfall and route capacity, test plausible timing scenarios and identify locations needing organiser attention. Forecast queue locations and ranges under alternative arrival assumptions without identifying people.

Answer boundary

The result is an estimate, not a crowd-safety certification or instruction to control people. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Supports the lecturer to brief the organiser on likely pressure points and adjust arrival communication before the event.

LT-054

HIGH | L3 - opening-time recommendation | R3 - research-grade or restricted

Lecturer/tutor question: Should I ask the event team for earlier room access because arrivals are concentrated or the room needs recovery time?

Sensors and telemetry

Proposed evidence: room CO_2, temperature/RH and anonymous occupancy; doorway/entrance flow; authorised room, AV and HVAC status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: event booking, permitted access/opening window, expected attendance, current room use, cleaning/maintenance, HVAC guidance, staffing and security arrangements. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: arrival and environmental forecast plus recommendation. Estimate arrival concentration and room recovery/readiness, then identify whether an earlier authorised opening would plausibly help and by how much. Recommend an earlier opening only to the authorised event owner, with uncertainty and a recheck time.

Answer boundary

The service cannot change opening times, staffing, security or access. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Provides the lecturer with a reasoned request to the event team rather than an unsupported preference.

6. Visiting speakers, events and irregular teaching

Questions LT-055-LT-056

LT-055

MEDIUM | L3 - acoustic suitability | R3 - research-grade or restricted

Lecturer/tutor question: Can the room support audience questions clearly using only numerical acoustic evidence, without recording speech?

Sensors and telemetry

Proposed evidence: numerical background sound, reverberation or speech-transmission metrics from commissioned acoustic tests or privacy-preserving sensors; authorised assistive-listening and microphone status. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection; numerical acoustic or flicker metrics only; no audio, speech or images. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room layout/capacity, installed microphone and assistive systems, event format, accessibility requirements and maintenance status. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: acoustic suitability assessment. Use commissioned numerical room-acoustic indicators, PA status and current layout; where available, use validated speech-transmission metrics. Retain no audio and do not infer audience behaviour.

Answer boundary

Numerical acoustic evidence cannot guarantee intelligibility for every listener and must not record, transcribe or analyse speech. If the installed sound/assistive system or representative acoustic coverage is unverified, provide no room-suitability conclusion and defer to AV/accessibility support.

Why this is useful: Supports a room or microphone check for audience participation without recording or analysing speech.

LT-056

CORE | L2 - evening event constraint check | R2 - integration-dependent

Lecturer/tutor question: Is this evening session likely to conflict with closure, cleaning, maintenance, security or reduced support?

Sensors and telemetry

Authorised status for access doors, lift, room power, AV and network; proposed room environment for current context only. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: event/timetable booking, building opening and closure, cleaning and maintenance schedule, after-hours access, support hours, security arrangements and room owner approval. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: authorised constraint lookup and readiness recommendation. Identify direct conflicts, service cut-off times and conditional items; propose a recheck or approved contingency where evidence supports one. Report confirmed conflicts and unknowns; do not infer permission from normal telemetry.

Answer boundary

Do not override closure, cleaning, maintenance or access rules. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Prevents a late session from failing because a non-teaching service or access condition was overlooked.

6. Visiting speakers, events and irregular teaching

Questions LT-057-LT-058

LT-057

HIGH | L3 - contingency planning | R2 - integration-dependent

Lecturer/tutor question: What lecturer contingency should I prepare if the lift or main AV setup becomes unavailable before the event?

Sensors and telemetry

Authorised live lift, door, AV, room-computer, power and network status; proposed route footfall and room environment only after candidate options are verified. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current event room, accessible routes, backup-room eligibility, booking, access, equipment, support and institutional contingency/accessibility guidance. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: scenario-based constrained recommendation. Build separate lift-loss and AV-loss cases, apply accessibility and room-authority constraints, and provide lecturer-owned communication or teaching fallbacks with decision deadlines. Show trigger, decision deadline, trade-offs and responsible service.

Answer boundary

The service cannot relocate the event, grant access, operate equipment or redefine an accessible arrangement. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the recommendation not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Allows the lecturer to brief speakers and attendees on an approved fallback before a single failure causes confusion.

LT-058

MEDIUM | L3 - post-event recovery | R2 - integration-dependent

Lecturer/tutor question: After a reception or large event, when are conditions likely to recover, and when is the room formally released for teaching?

Sensors and telemetry

Proposed evidence: room CO2, temperature/RH, PM2.5 and anonymous occupancy, with HVAC airflow/state; event-related TVOC only where justified. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: event end, next booking, cleaning/room-release status, HVAC or room-owner guidance, maintenance and any approved clearance requirement. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: recovery observation and forecast. Detect event-related peaks, compare decay with the verified baseline, estimate a recheck time and keep formal room release separate from environmental recovery. Forecast return with uncertainty and a recheck time, separate from official room release.

Answer boundary

Environmental readings cannot authorise reoccupation or certify cleanliness/safety. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Helps the lecturer distinguish likely environmental recovery from the room owner's formal release decision.

6. Visiting speakers, events and irregular teaching

Questions LT-059-LT-060

LT-059

CORE | L2 - guest connectivity guidance | R2 - integration-dependent

Lecturer/tutor question: Which approved network, presentation and account requirements should I send an external speaker before arrival?

Sensors and telemetry

No environmental sensing is required. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: visitor/network policy, approved guest access process, room equipment, supported file/interface options, accessibility guidance, support contact and current incidents. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: authorised policy and capability lookup. Match the speaker's minimal stated needs to approved options, provide setup instructions and an offline or wired fallback where authorised. Provide authorised setup steps and a wired/offline fallback.

Answer boundary

Do not create accounts, promise compatibility, request unnecessary device details or recommend bypassing security. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Gives the speaker clear preparation instructions and reduces avoidable setup time on arrival.

LT-060

MEDIUM | L2 - shareable evidence summary | R2 - integration-dependent

Lecturer/tutor question: What non-sensitive, authorised evidence may I share with the organiser about the room's current suitability?

Sensors and telemetry

Use quality-controlled aggregate room environment, capacity-relevant anonymous occupancy history and authorised service-readiness summaries where disclosure is permitted. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room purpose/capacity, accessibility features, equipment inventory, booking status, approved public information and data-sharing permissions. Current event, visitor, room, access, support and safety records, with effective/retrieved time and authorised disclosure scope, take precedence over telemetry or historical plans.

Analysis required

Operation: evidence-package compilation. Include source owner, date, spatial coverage, method, uncertainty and current-status caveats; separate verified room features from observed conditions and forecasts. Label each item as lookup, observation or calculation, with date, permission and limitation.

Answer boundary

Do not present the package as a booking, accessibility or safety certificate, disclose restricted system details or use historic room functions as current facts. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the event, visitor, room or safety owner.

Why this is useful: Supports the lecturer to give the organiser a concise, defensible room summary rather than unsupported reassurance.

7. Inclusive teaching, well-being, safety and emergencies

Questions LT-061-LT-062

LT-061

HIGH | L3 - fatigue-risk planning | R2 - integration-dependent

Lecturer/tutor question: For this three-hour class, should I plan additional breaks or activity changes because of the room and session conditions?

Sensors and telemetry

Proposed evidence: room CO_2, temperature/RH, illuminance and numerical sound, with anonymous occupancy and verified airflow context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: session duration, expected attendance, room capacity/layout, accessibility requirements, approved teaching schedule and current HVAC incidents. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: long-session forecast and teaching recommendation. Estimate environmental change and variability, identify likely recheck points, and suggest flexible breaks or activity changes while keeping each evidence type separate. Recommend breaks or activity variation as teaching choices, not health or performance judgements.

Answer boundary

Do not infer fatigue, health, engagement or individual need from sensors, and do not prescribe a universal break interval. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the forecast not assessable and defer to the accessibility, security, first-aid or safety authority.

Why this is useful: Supports the lecturer to plan a sustainable session structure before conditions or concentration concerns emerge.

LT-062

CORE | L2 - verified accessibility check | R2 - integration-dependent

Lecturer/tutor question: Before class, which step-free route, entrance, lift and accessible-seating information is currently confirmed?

Sensors and telemetry

Use authoritative current lift and door status; proposed anonymous route-flow bands may describe congestion only. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current accessible-route map, entrance and room access, designated seating/layout, temporary closures, lift service, room assignment and accessibility guidance. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: authorised accessibility lookup and route calculation. Verify each requested feature, state effective times and give an approved fallback route or contact where a component is unavailable. Optional aggregate flow may describe congestion; it cannot certify accessibility or replace competent advice.

Answer boundary

Do not infer a person's needs, certify the route for every user or substitute occupancy/environmental data for accessibility information. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the calculation or estimate not assessable and defer to the accessibility, security, first-aid or safety authority.

Why this is useful: Allows the lecturer to give accurate pre-class information and plan an inclusive arrival rather than relying on memory.

7. Inclusive teaching, well-being, safety and emergencies

Questions LT-063-LT-064

LT-063

HIGH | L2 - hearing-access readiness | R2 - integration-dependent

Lecturer/tutor question: For students using hearing support, are the authorised assistive-listening system and relevant room sound conditions ready?

Sensors and telemetry

Authorised assistive-listening and microphone health; proposed numerical background sound/reverberation indicators with no audio retention. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room inventory, receiver/loan procedure, accessibility guidance, room layout, session assignment and support contact. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: authorised readiness lookup plus acoustic context observation. Verify the current assistive-listening inventory, authorised test and maintenance status, then add numerical room-acoustic context. Report unknowns and approved support contact without guaranteeing hearing outcomes.

Answer boundary

Do not infer who uses hearing support, record speech or guarantee suitability for every student. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the observation not assessable and defer to the accessibility, security, first-aid or safety authority.

Why this is useful: Supports the lecturer to verify an important teaching adjustment before the session starts.

LT-064

MEDIUM | L3 - lighting-stability assessment | R3 - research-grade or restricted

Lecturer/tutor question: Is the lighting stable enough for students sensitive to glare or flicker, and which areas have not been measured?

Sensors and telemetry

Proposed evidence: illuminance and, where justified, validated flicker measurements at representative occupied and presentation zones; daylight/solar and lighting-state context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: room lighting design/controls, current layout, window/blind state where available, maintenance incidents, accessibility guidance and approved teaching use. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: lighting-stability assessment. Quantify variation, flicker metric where valid and likely glare conditions over the teaching period; distinguish direct measurements from estimates and identify controllable versus fixed factors. Report variability, daylight/confounders, coverage and uncertainty.

Answer boundary

Ordinary lux sensors cannot establish flicker, and telemetry cannot certify suitability for an individual. If no commissioned flicker measurement or authoritative lighting/accessibility evidence exists, report illuminance only and mark flicker suitability not assessable.

Why this is useful: Supports an inclusive lighting adjustment or support request based on the evidence actually available.

7. Inclusive teaching, well-being, safety and emergencies

Questions LT-065-LT-066

LT-065

CORE | L2 - occupancy versus capacity | R2 - integration-dependent

Lecturer/tutor question: Is the room becoming too crowded for the planned activity and accessible circulation?

Sensors and telemetry

Proposed evidence: anonymous room occupancy/doorway counts and zone-level presence, with known count error and current timestamps. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room capacity, layout, accessible routes/seating, activity plan, booking and relevant safety guidance. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: current occupancy estimate plus authoritative suitability check. Compare the occupancy range and planned activity with the approved configuration; identify evidence of constrained circulation and a lecturer-owned adjustment. Report uncertainty and do not make a safety certification.

Answer boundary

Do not identify attendees, claim exact attendance or certify safety. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the accessibility, security, first-aid or safety authority.

Why this is useful: Supports the lecturer to pause, simplify or seek support before the activity makes movement difficult for students.

LT-066

CORE | L2 - high-CO2 response guidance | R2 - integration-dependent

Lecturer/tutor question: If the room's CO_2 indicator stays elevated, which lecturer actions are authorised by current guidance?

Sensors and telemetry

Proposed evidence: direct NDIR CO_2 at representative room points, temperature/RH, anonymous occupancy and verified airflow/HVAC state. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room/session, approved ventilation response guidance, active HVAC incidents, operable-opening rules, support contact and emergency policy. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: current observation and policy-bounded recommendation. Confirm persistence and source agreement, check authorised ventilation/service status, then present the approved lecturer actions and a recheck/escalation condition. Present bounded lecturer actions and recheck timing, not a safety or medical judgement.

Answer boundary

CO_2 is a ventilation indicator, not a complete air-quality or safety measure. Do not invent thresholds, override controls or give medical advice; if current guidance, room coverage or HVAC context is missing, provide the reading and defer action to the responsible authority.

Why this is useful: Gives the lecturer an authorised response path while keeping ventilation, health and safety judgement with the responsible authority.

7. Inclusive teaching, well-being, safety and emergencies

Questions LT-067-LT-068

LT-067

CORE | L1 - authoritative emergency information | R3 - research-grade or restricted

Lecturer/tutor question: If an alarm activates, can the service relay only the official status and instructions, with the alarm system taking precedence?

Sensors and telemetry

Use only authorised certified alarm, public-address and official emergency-message status. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: certified alarm system, official emergency messaging, building emergency plan, security instructions and current route/closure information where released for occupants. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: authoritative emergency lookup and presentation. Display the latest official instruction, source and timestamp, prioritise certified information and clearly mark any unavailable or supplementary data. Do not combine experimental telemetry into an alternative instruction or recommend re-entry.

Answer boundary

Certified alarms and official instructions always take precedence. The conversational service must not reinterpret, delay, suppress or personalise an alarm; if the official feed is unavailable or conflicting, direct users to follow the alarm and current emergency procedure.

Why this is useful: Provides a clear route to the same authoritative information without creating a competing emergency channel.

LT-068

HIGH | L2 - after-hours permission lookup | R3 - research-grade or restricted

Lecturer/tutor question: I have a small evening tutorial. Is the session authorised, and which after-hours or lone-working procedure applies?

Sensors and telemetry

Authorised door, lift and service status may describe current availability; proposed anonymous occupancy is not permission and must not be used to identify the group. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current booking, building opening, staff/student access, after-hours teaching approval, lone-working/check-in procedure, security contact, room eligibility and closures. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: authorised policy and permission lookup. Confirm whether the session is approved, list required check-in or communication steps and identify service limitations or a decision deadline. State permission, required check-in steps and responsible contact; normal telemetry cannot approve the session.

Answer boundary

The service cannot grant after-hours access, waive lone-working rules, identify attendees or treat normal building telemetry as approval. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the accessibility, security, first-aid or safety authority.

Why this is useful: Helps the tutor arrange evening teaching responsibly before arriving with students to a closed or unsupported building.

7. Inclusive teaching, well-being, safety and emergencies

Questions LT-069-LT-070

LT-069

CORE | L2 - temporary barrier notification | R2 - integration-dependent

Lecturer/tutor question: Can you warn me before class if a verified lift, door or route change creates an accessibility barrier?

Sensors and telemetry

Authorised lift and door state plus proposed aggregate route footfall; no user tracking. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room assignment, accessible-route graph, temporary closure/maintenance notices, access restrictions and approved accessibility communication channels. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: scheduled authoritative lookup and barrier notification. Re-evaluate the verified route at a defined pre-class time, identify the affected segment and provide an authorised alternative or contact. Notify only verified barriers, give an approved alternative and avoid tracking any individual user.

Answer boundary

Do not infer who needs the route, certify an alternative for every user or suppress official notices. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the accessibility, security, first-aid or safety authority.

Why this is useful: Supports the lecturer to communicate a barrier early and avoid students discovering it only when they arrive.

LT-070

MEDIUM | L2 - aggregate comfort feedback | R2 - integration-dependent

Lecturer/tutor question: Can students submit anonymous comfort reports, and can I see only an aggregate pattern during the class?

Sensors and telemetry

Optional anonymous comfort reports plus proposed room CO_2, temperature/RH, PM2.5 and numerical sound/light context. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; numerical acoustic or flicker metrics only; no audio, speech or images. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: approved feedback tool, consent/privacy notice, minimum group size, current room/session and support/escalation policy. Official accessibility, safety, lone-working and emergency sources, with effective time and competent authority, take precedence over telemetry.

Analysis required

Operation: privacy-preserving aggregation and trend comparison. Summarise report counts/categories only above the approved threshold, compare with room observations and state whether evidence converges or conflicts. Compare with room evidence and historical baseline; report sample size, bias and uncertainty without individual inference.

Answer boundary

Accept voluntary anonymous reports only; do not identify respondents, infer health or engagement, or expose small groups. If privacy thresholds, consent, room evidence or moderation rules are unmet, withhold the aggregate pattern and provide the approved reporting route.

Why this is useful: Shows whether several anonymous reports align with room evidence without profiling students or interpreting individual health.

8. Reliable services, sustainability and evidence transparency

Questions LT-071-LT-072

LT-071

MEDIUM | L3 - constrained sustainability recommendation | R2 - integration-dependent

Lecturer/tutor question: Is this recurring class using substantially more space than it needs, and which lower-energy eligible room could I request next time?

Sensors and telemetry

Proposed evidence: anonymous occupancy, target-room environment and valid room-level or end-use HVAC/lighting/plug energy; verified BMS state. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: approved capacity, current eligible rooms, timetable/booking constraints, accessibility, equipment, permitted teaching use and room-change process. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: utilisation calculation, matched energy comparison and constrained recommendation. Compare equivalent sessions, preserve capacity/access/equipment constraints and provide a candidate for a future request with uncertainty and trade-offs. Report savings range, comfort/accessibility trade-offs and a future room-request option.

Answer boundary

Do not infer attendance or teaching quality, assume an alternative is available, or make a room claim from whole-building energy. Without current eligibility, accessibility, equipment and mapped room-energy evidence, provide no lower-energy room recommendation.

Why this is useful: Supports the lecturer to make a proportionate future room request that considers both teaching requirements and avoidable energy.

LT-072

MEDIUM | L2 - action-effect calculation | R2 - integration-dependent

Lecturer/tutor question: Did switching off permitted lights and teaching equipment after class measurably reduce energy without affecting the next session?

Sensors and telemetry

Authorised lighting/equipment state, correctly mapped circuit power and energy, illuminance and next-session service readiness. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current shutdown guidance, session and next-booking times, installed equipment inventory, mandatory always-on services and maintenance incidents. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: calculation and matched comparison. Estimate energy avoided against comparable post-class periods, verify required systems remained ready and report a plausible range rather than false precision. Report baseline, duration, confounders and uncertainty; do not claim causality from one event.

Answer boundary

Do not encourage switching off required, safety-critical or room-owner-controlled equipment. If asset-level metering, approved shutdown rules or a comparable baseline is missing, report only the authorised action and mark the energy effect not assessable.

Why this is useful: Shows whether a simple lecturer-owned behaviour has a real benefit without compromising the next class.

8. Reliable services, sustainability and evidence transparency

Questions LT-073-LT-074

LT-073

MEDIUM | L3 - multi-outcome room comparison | R3 - research-grade or restricted

Lecturer/tutor question: For a future room request, how do two eligible rooms compare on teaching fit, accessibility, environment, service reliability and energy?

Sensors and telemetry

Proposed evidence: comparable room environment and anonymous occupancy; valid end-use energy/BMS evidence; authorised aggregate AV/network service history. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection; anonymous occupancy with count uncertainty or blind spots shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room eligibility, capacity, accessibility, equipment, permitted use, bookings and de-identified service records. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: matched multi-outcome comparison and recommendation. Normalise by session type, duration, occupancy band and weather; present each outcome separately, then rank only using lecturer-selected priorities and show missing criteria. Show normalisation, trade-offs, sensitivity and uncertainty rather than one opaque score.

Answer boundary

Do not collapse unlike evidence into an unexplained score, guarantee future conditions or recommend an ineligible room. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Supports a balanced room request instead of optimising energy, technology or comfort in isolation.

LT-074

HIGH | L3 - recurring disruption analysis | R3 - research-grade or restricted

Lecturer/tutor question: Which confirmed room or service problems have recurred across my teaching sessions, and how often?

Sensors and telemetry

Authorised AV/IT/BMS/power/lift incident events and quality-controlled room environment, linked only to session windows. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: lecturer's session times/rooms, de-identified incident and maintenance logs, room-to-service mapping, resolution status and configuration changes. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: event matching and recurrence analysis. Link confirmed incidents to sessions, group by failure domain, quantify frequency/duration and distinguish verified disruption, correlated anomaly and unreported inconvenience. Count recurrence, duration and affected services, account for configuration changes and report evidence gaps.

Answer boundary

Do not attribute blame, infer teaching impact beyond the recorded evidence or expose restricted operational details. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Provides the lecturer with a factual basis for escalating persistent problems and planning around known recurring risks.

8. Reliable services, sustainability and evidence transparency

Questions LT-075-LT-076

LT-075

CORE | L2 - sensor trust assessment | R1 - practical

Lecturer/tutor question: Is this sensor reading reliable and spatially representative enough for the teaching decision I am making?

Sensors and telemetry

Sensor identity, observed property, exact room/zone mapping, stated accuracy, commissioning/calibration, sampling/aggregation, last contact, uptime, gaps, drift, relocation and agreement with redundant or reference points. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current sensor inventory, maintenance record, room mapping, data-quality policy and any active fault or calibration notice. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: sensor-quality assessment. Check freshness, valid range, persistence, spatial suitability, redundancy, drift and maintenance; classify the reading as usable, usable with caution or not usable for the stated decision. Return a qualified usability judgement.

Answer boundary

Do not hide disagreement, substitute a corridor sensor or present an indicative low-cost device as certified evidence. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Prevents a teaching decision from resting on stale, misplaced or poorly commissioned sensing.

LT-076

HIGH | L3 - source reconciliation | R2 - integration-dependent

Lecturer/tutor question: Why does the talking-building answer disagree with the wall display or another dashboard?

Sensors and telemetry

For each displayed value, use source provenance: sensor or derived metric, exact location, unit, aggregation, timestamp, last refresh, calibration/health and processing version. Use source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current data catalogue, room/sensor mapping, dashboard configuration, threshold/unit definitions, cache policy and maintenance/change history. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: provenance comparison and source reconciliation. Align location, units and time windows; test stale cache, different sensors, aggregation or derived definitions; identify the authoritative source for the specific question. Explain definitional, refresh or lineage differences and leave unresolved conflicts explicit.

Answer boundary

Do not choose the value that merely looks plausible or conceal unresolved lineage. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Prevents the lecturer from acting on two values that differ because they represent different places, times or definitions.

8. Reliable services, sustainability and evidence transparency

Questions LT-077-LT-078

LT-077

CORE | L2 - claim-type transparency | R1 - practical

Lecturer/tutor question: Can you label each material part of your answer as observed, looked up, calculated, estimated, forecast, diagnosed or recommended?

Sensors and telemetry

No new sensor is required. Contextual telemetry, if used, must be current, permissioned and correctly mapped; it cannot substitute for the named authoritative record.

Authoritative supporting data

Authoritative source owner, observed/effective/retrieved time, spatial scope, permissions and method/version for each derived result. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: evidence-type explanation. Label every material statement, list inputs and time window, show uncertainty and separate the direct answer from assumptions, diagnosis and recommendation. Show inputs, spatial/time scope, method/version, uncertainty and omitted evidence.

Answer boundary

Do not use fluent text to disguise inference, forecast or missing evidence. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the diagnosis not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Makes the evidential status of the answer visible so the lecturer can judge how much reliance to place on it.

LT-078

CORE | L1 - answerability check | R1 - practical

Lecturer/tutor question: There is no verified sensor in this room. Which facts can you still confirm, and which conditions are not assessable?

Sensors and telemetry

Use the current sensor inventory, exact spatial coverage and room/HVAC mapping. Use 1-minute aggregates from commissioned points; live values no more than 5 minutes old and historical windows at least 90% complete; anonymous route-flow bands and current authorised lift/door status, with count error and uncovered segments shown. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current room identity, purpose, capacity, access, booking, AV/service and HVAC-zone mapping, with current incident or maintenance status. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: answerability assessment. Identify which requested variables lack valid room coverage, return authoritative room/service facts separately, and explain what measurement or manual check would be needed. Return authoritative room/service facts separately; do not substitute corridor or adjacent-room measurements for a room value.

Answer boundary

Do not interpolate a precise room value, substitute nearby sensing or claim comfort/air quality. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the lookup or analytical result not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Preserves trust by confirming what is known while refusing to invent room conditions from nearby sensors.

8. Reliable services, sustainability and evidence transparency

Questions LT-079-LT-080

LT-079

MEDIUM | L3 - control-strategy recommendation | R3 - research-grade or restricted

Lecturer/tutor question: For this recurring class, is there an efficiency change I could ask the authorised team to evaluate without weakening comfort, air quality or accessibility?

Sensors and telemetry

Proposed evidence: room environment, anonymous occupancy, illuminance, lighting state, HVAC airflow/valve/damper state and correctly mapped energy submetering. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: current timetable, room use, control policy, visual/accessibility requirements, maintenance state and the approved BMS/energy review process. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: historical comparison, bounded scenario calculation and recommendation for expert review. Identify avoidable operation under comparable sessions, estimate energy and environmental trade-offs and state uncertainty. State assumptions, trade-offs and uncertainty and route any control proposal to expert review.

Answer boundary

The service cannot change controls, weaken air-quality, comfort, visual or accessibility requirements, or certify savings. Without matched room, occupancy, BMS and energy evidence, present no efficiency recommendation and defer any control change to expert review.

Why this is useful: Supports the lecturer to contribute recurring schedule knowledge to a responsible efficiency discussion without taking over building control.

LT-080

HIGH | L3 - end-of-term evidence synthesis | R3 - research-grade or restricted

Lecturer/tutor question: At the end of term, which verified room and service patterns should inform how I plan these sessions next time?

Sensors and telemetry

Quality-controlled session-level room environment, anonymous occupancy bands, numerical sound/light, authorised aggregate AV/network incidents, lift/access disruptions and valid energy context. Use room/asset-mapped 1-minute demand and 15-minute energy with a stated allocation method; source-timestamped authorised asset/service events, with no teaching, device or network-content inspection. Report spatial/asset mapping, clock alignment, commissioning/calibration, uptime, gaps, drift and maintenance state.

Authoritative supporting data

Authoritative data: historical session times/rooms, room changes, current room/service definitions, maintenance/closure events and future-planning constraints. Current room, service, energy and provenance records, with effective/retrieved time, source owner and permission, take precedence over inferred status and the historical plan.

Analysis required

Operation: longitudinal evidence synthesis and lecturer-owned recommendation. Summarise recurring room/time patterns, matched comparisons, confirmed service disruptions, uncertainty and evidence gaps; identify practical planning questions for responsible teams. Summarise recurring patterns, evidence gaps, uncertainty and lecturer-owned planning choices.

Answer boundary

Do not infer student engagement, learning, lecturer performance or causality from correlations, and do not present historical room use as current. If a hard-constraint source or required room/asset evidence is stale, missing, conflicting, below quality threshold or unauthorised, return confirmed facts only, mark the comparison not assessable and defer to the room, energy, maintenance or data owner.

Why this is useful: Turns term-level evidence into concrete preparation and room-request lessons without evaluating students or staff.

4. Catalogue summary and implementation requirements

The catalogue combines practical teaching questions with integration-dependent room, access, AV and timetable reasoning, plus research-grade comparison, diagnosis and safety-sensitive cases. Readiness identifies the verified sensing, current integrations, source authority, spatial and temporal coverage, quality threshold, permissions and safe-failure controls required for a responsible answer.

Questions	80	Eight lecturer/tutor-centred sections containing ten questions each.
R1 practical	4	Can be supported with proposed core sensing and basic current spatial, service or provenance feeds.
R2 integration-dependent	54	Require timetable, room, booking, access, accessibility, AV/IT, BMS, maintenance, visitor or forecasting integration.
R3 research or restricted	22	Require dense coverage, validated comparison/diagnosis, controlled operational data, or stronger privacy and safety controls.

Recurring system capabilities

1 Teaching-day planning

Resolve session preparation, routes, office hours, visitors, hybrid delivery, breaks, contingencies and irregular-hour constraints.

2 Room, access and permission reasoning

Use current assignments, room purpose, capacity, access, bookability, accessibility, visitor rules, routes, lift status and closures.

3 Environmental, AV and service evidence

Combine correctly located room sensing with anonymous occupancy, BMS, power, AV, network, assistive technology and authorised service status.

4 Forecast, comparison and diagnosis

Predict conditions and flow, build matched baselines, compare eligible rooms, reconstruct service events and avoid unsupported causal claims.

5 Evidence quality and provenance

Check room/asset mapping, sampling, clocks, commissioning/calibration, uptime, gaps, drift, source agreement, processing lineage and permissions.

6 Privacy, accessibility, safety and abstention

Minimise personal and teaching data, refuse unauthorised access or safety claims, preserve inclusive alternatives and return not assessable when evidence is inadequate.

QUESTION RECORD STRUCTURE

Each entry can become a system test containing the natural utterance and paraphrases, required entities, minimum room and service evidence, source-precedence rule, quality threshold, expected operation type, permission rule, answerability class, acceptable clarification, uncertainty statement, safe failure, expected answer characteristics and prohibited overclaims.

5. Evidence, permissions and answerability rules

These rules apply across all 80 records and define when the building may answer, qualify, ask for clarification, defer to an authorised service or return not assessable.

Building and room status

The May 2021 plan is a historical spatial reference. Current room purpose, ownership, capacity, staff access, bookability, equipment, accessible features, entrances, lifts, services and closures require an authoritative current source.

Spatial coverage

A room-level answer requires a correctly located sensor inside that room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

Time window and operation type

Every answer resolves the requested interval and labels its operation as observation, authorised lookup, calculation, estimate, forecast, comparison, diagnosis or recommendation. Preserve observed-at and retrieved-at times.

Calibration and data quality

Use stated sampling and aggregation, synchronised clocks, commissioning/calibration, uptime, gap and drift flags, maintenance state and source agreement. Failed, stale, relocated or unverified points cannot support a precise claim.

Teaching-content minimisation

Do not inspect teaching materials, recordings, screens, speech, conversations, network traffic or device content. Process only the minimum schedule, activity and requirements needed for the request.

Occupancy, sound and privacy

Use anonymous aggregate counts and numerical LAeq or percentile statistics. Do not identify people, retain speech, infer conversation or attendance, track movement or estimate engagement, learning or teaching performance.

Environmental interpretation

Direct-reading NDIR CO_2 is a ventilation indicator, not complete air quality. Temperature, RH, PM, indicative TVOC, airflow, radiant conditions, light, outdoor weather and activity may be relevant subject to coverage and quality.

Teaching rooms and specialist activity

Environmental IoT data cannot authorise room use, practical activity or equipment operation, and cannot certify accessibility or safety. Current purpose, induction, risk controls and equipment status come from authoritative systems.

Operational services and permissions

Timetables, bookings, access, visitors, lifts, network, AV, capture, maintenance and closures come from authorised feeds. The service cannot book, grant access, operate plant or override responsible teams.

Emergency and high-consequence answers

Certified alarms and official communications take precedence. Safety, health, lone-working, accessibility and compliance questions require competent authority; missing or conflicting critical evidence requires a limited or not-assessable answer.

FINAL ANSWERABILITY RULE

Answer only when the required variables, current room or served-zone coverage, requested interval, source authority, quality threshold, permissions, accessibility and safety constraints are adequate. Otherwise ask for clarification, omit unsupported criteria, return confirmed facts with explicit uncertainty, restrict disclosure, defer to the responsible service, or mark the analytical part not assessable.